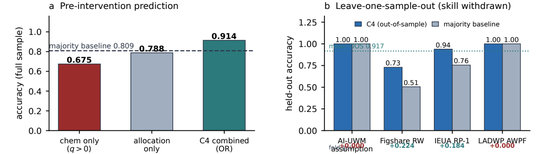

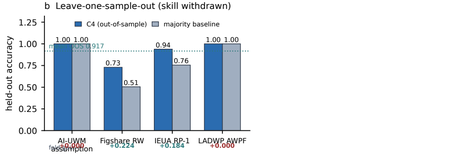

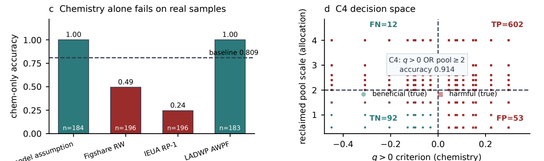

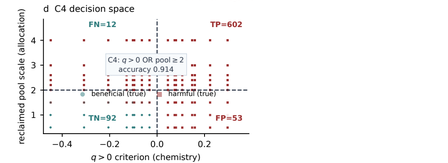

Data: A31_WP2/wp2_e3_independent_prediction_summary.json, wp2_e3b_combined_prediction_summary.json, wp2_e3_predictions.csv, wp2_e3b_features.csv; fold decomposition and the withdrawal ruling from A38_simulated-review/red_team_findings.md (finding F-2).
n_all = 813 reliable scenarios; n_eff = 733 after excluding 80 near-zero rows (|d5_pot| < 1e-9). Leave-one-sample-out groups: n=184/170/196/183; only the two complete-composition folds (IEUA 196, LADWP 183) may be used for sign claims.
LIMITATION: the out-of-sample gain (median +0.092; 0.917 vs 0.815) rests on only two transferable skill: it originates entirely in one of four folds (IEUA, +0.184), two folds are saturated by construction (baseline 1.000, gain 0.000) and one is negative (Figshare RW, -0.018); the median fold gain is 0.000 and removing the single positive fold makes the mean negative. The parameter grid is shared across folds, so only the chemistry axis is held out. With four folds and one effective fold this does not establish predictive skill, and the criterion is reported as an identified relationship under a stated data requirement - not as a validated predictor. AI-UWM and Figshare RW are incomplete-species samples, so their per-sample accuracies are supplementary-only.